



Engineering the F11 Journey.

GlobalIntern: A unified relational architecture for international student employment

The Unexpected Hurdle.



The Expectation

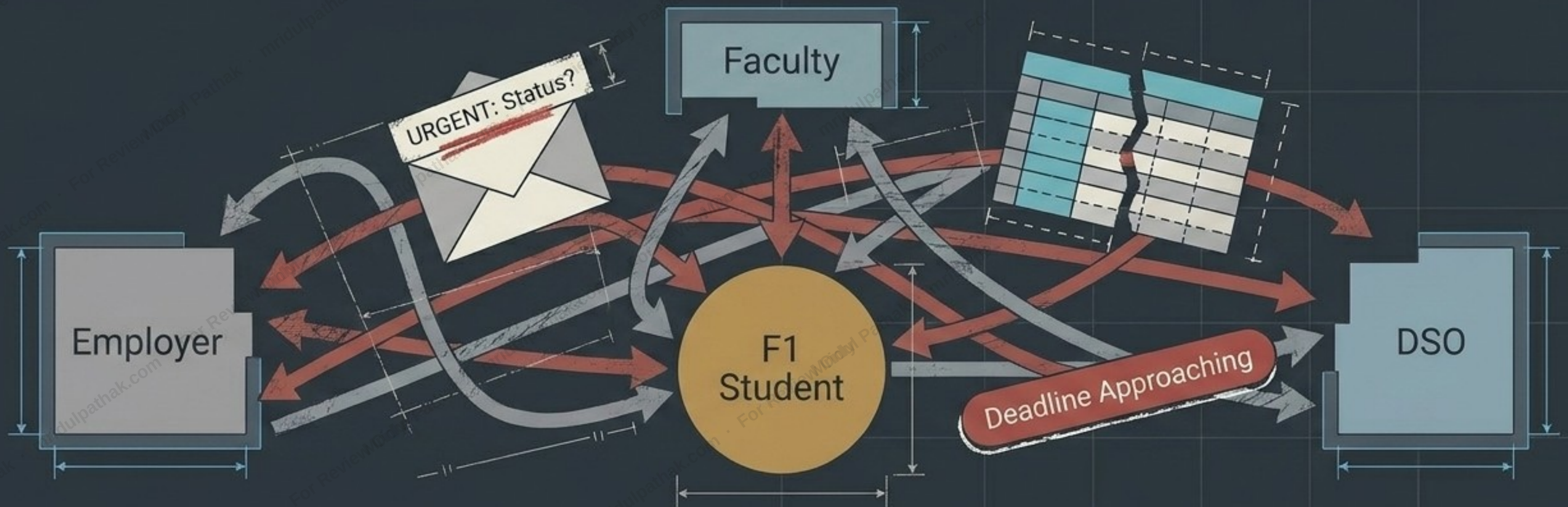
Mastering technical database design, passing rigorous coding rounds, and securing a competitive U.S. internship.

The Reality

Navigating the rigid, complex maze of Curricular Practical Training (CPT) work authorization before the start date expires.

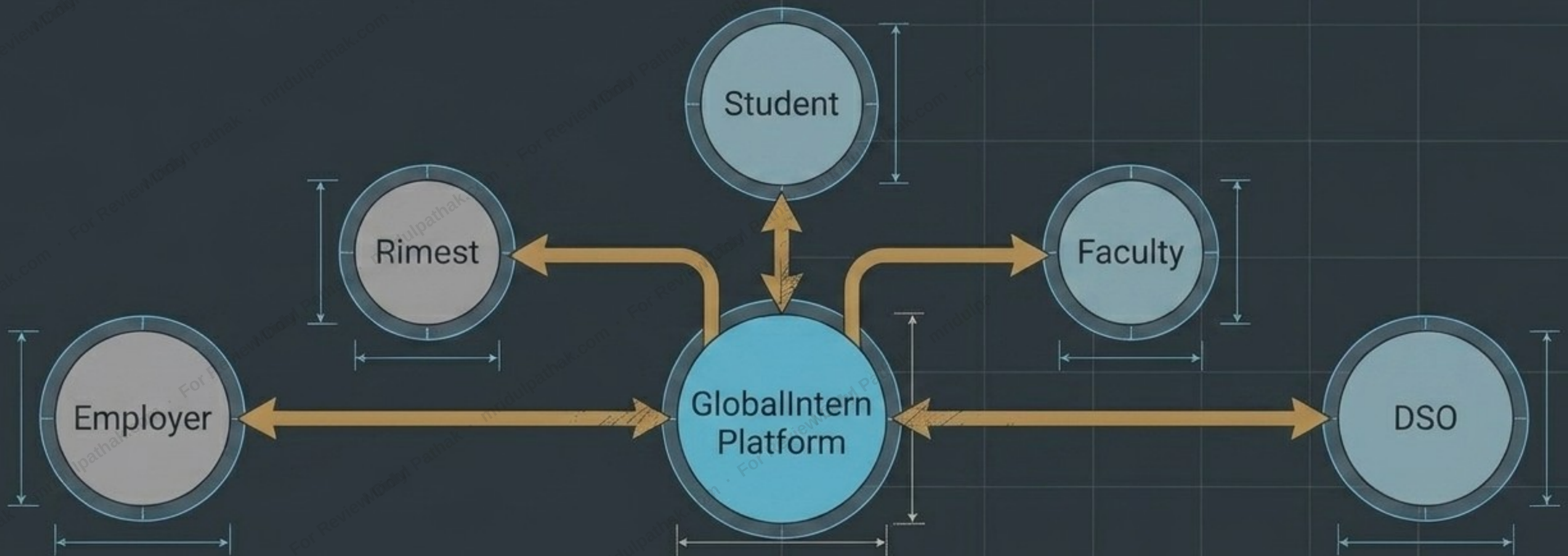
The Friction of Compliance.

Managing CPT requires flawless coordination between disconnected entities. When tracked manually via email chains and static spreadsheets, data is delayed, statuses are lost, and students risk losing their offers.



A Single Source of Truth.

GlobalIntern eliminates bureaucratic friction by centralizing the authorization lifecycle into a unified relational database.



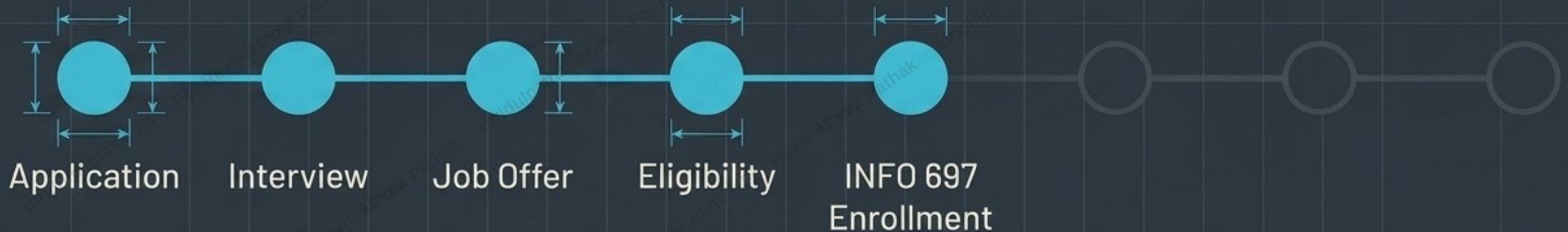
Phase 1: The Search.

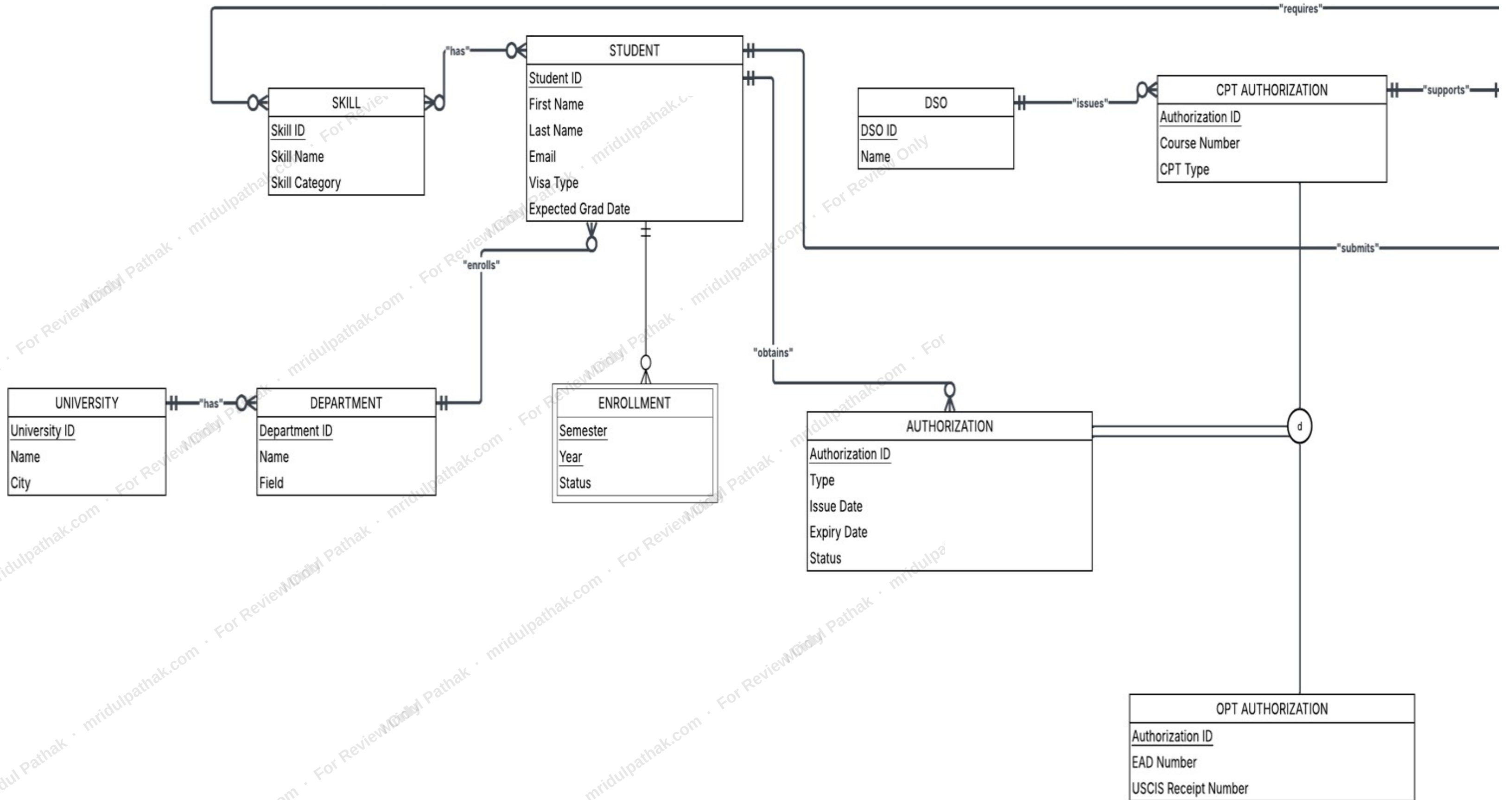
The student interacts entirely with the Employer entity. The database captures the initial pipeline, logging interview outcomes and formalizing the offer details (title, compensation, duration).



Phase 2: Academic Alignment.

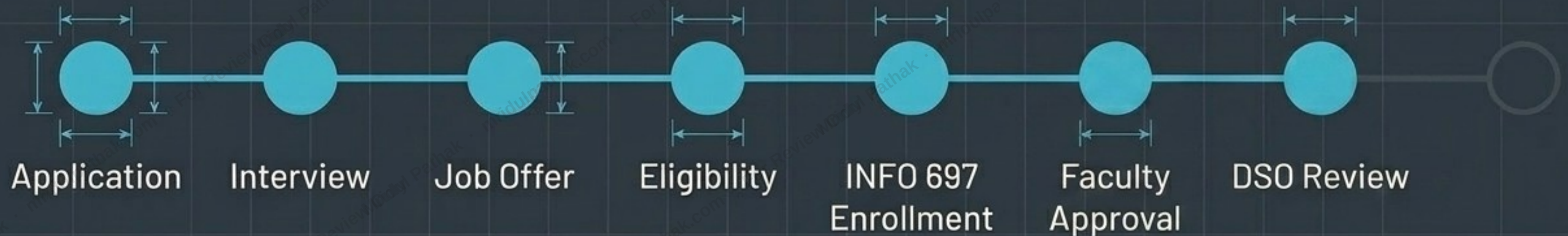
The system shifts focus to the University entity. The database verifies the student's expected graduation date and automatically links the specific internship to an active enrollment in the INFO 697 practicum course.

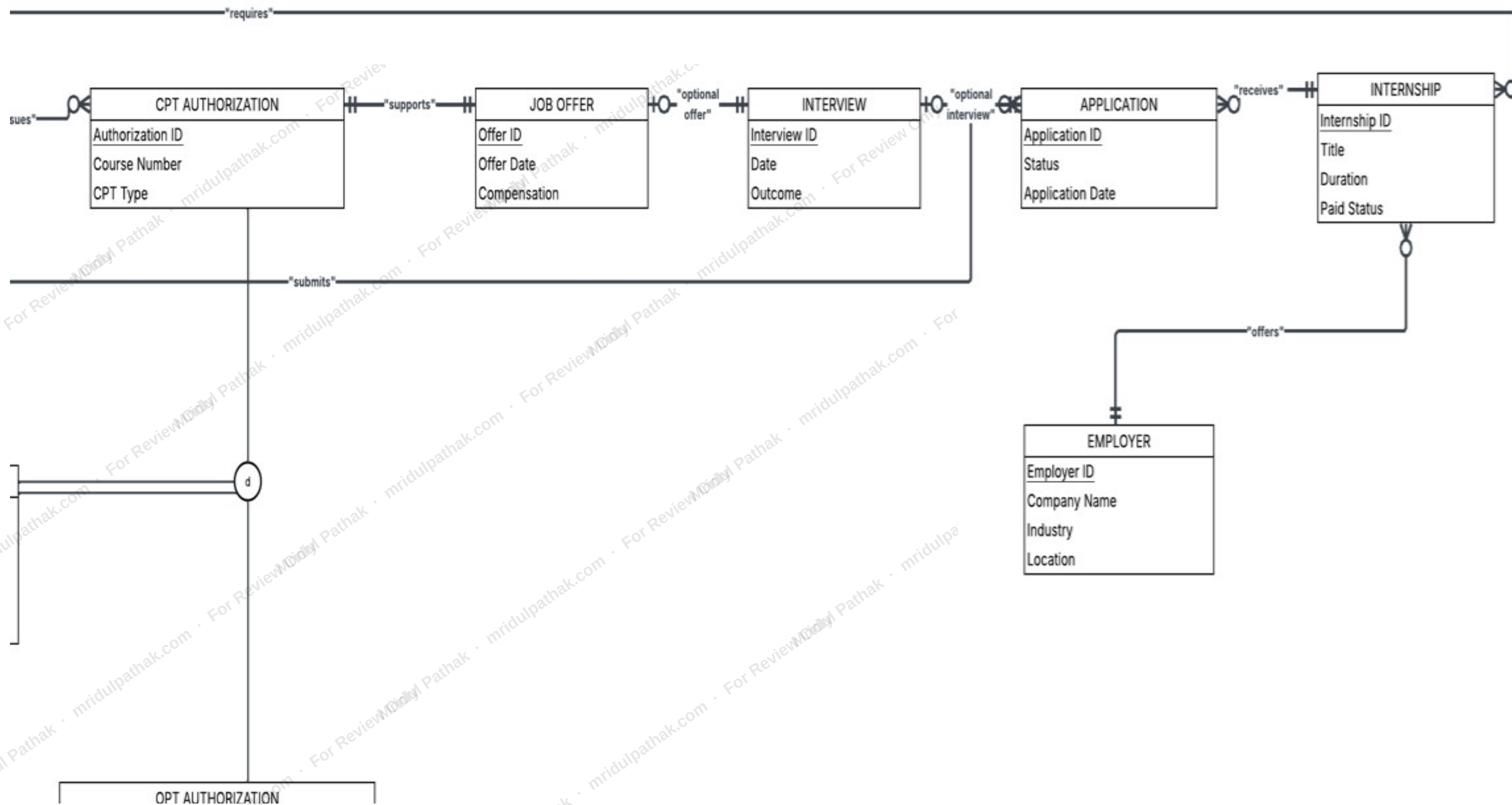




Phase 3: Compliance Clearance.

The Faculty and DSO (Designated School Official) entities are triggered. They review the aggregated data—no emails required. The DSO ensures federal compliance based on the verified INFO 697 enrollment and employer documentation.





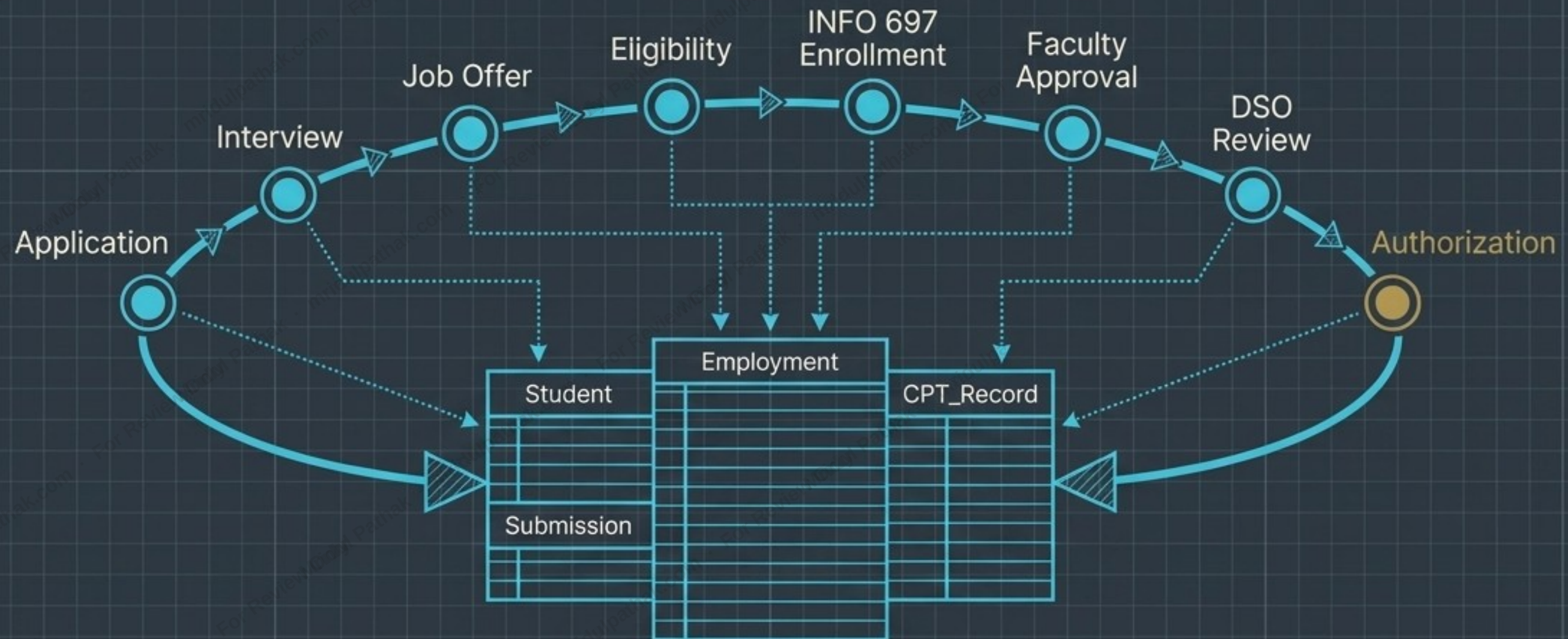
Phase 4: The Result.

The loop is closed. The student is officially authorized. Manual tracking is bypassed; the database now holds a permanent, queryable record of the exact CPT type, issue date, and expiry date.



The Journey is the Architecture

Every step in the bureaucratic timeline perfectly maps to a relational entity. The student's lived experience is governed by rigorous database logic.





APEX

App Builder ▾

SQL Workshop ▾

Team Development ▾

Gallery



↑ Object Browser

Schema

PATHAKM3 ▾



Tables ▾



APPLICATION

AUTHORIZATION

CPT_AUTHORIZATION

DEMONSTRATES

DEPARTMENT

DOCUMENT

DSO

EMPLOYER

ENROLLMENT

INTERNSHIP

INTERNSHIP_SKILL

INTERNSHIP_VISA

INTERVIEW

JOB_OFFER

OPT_AUTHORIZATION

SKILL

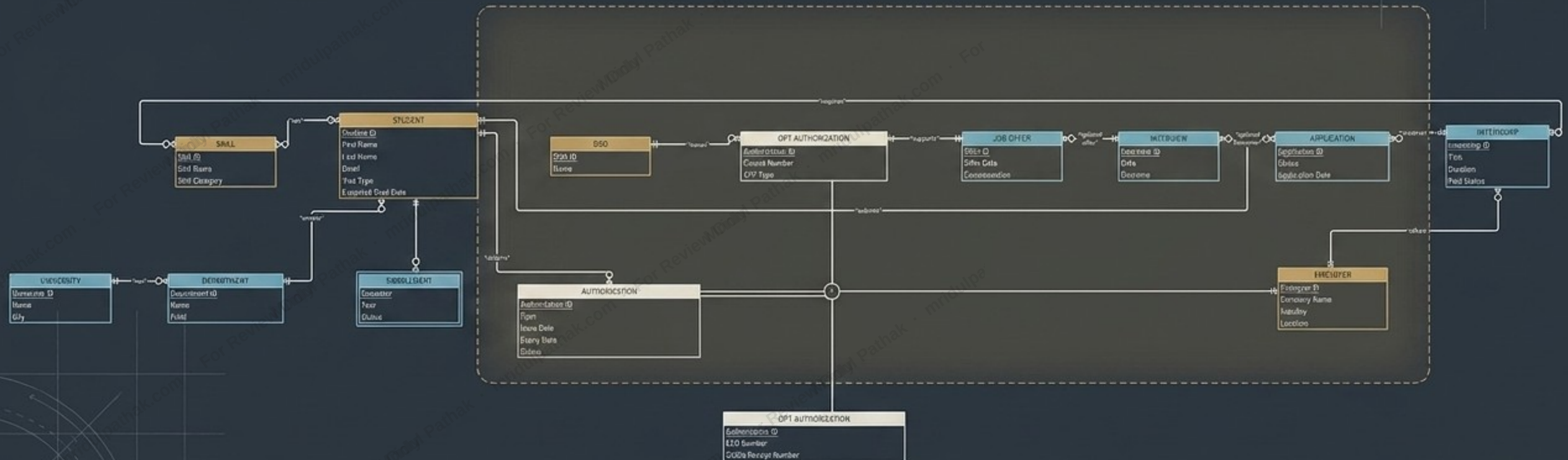
STUDENT

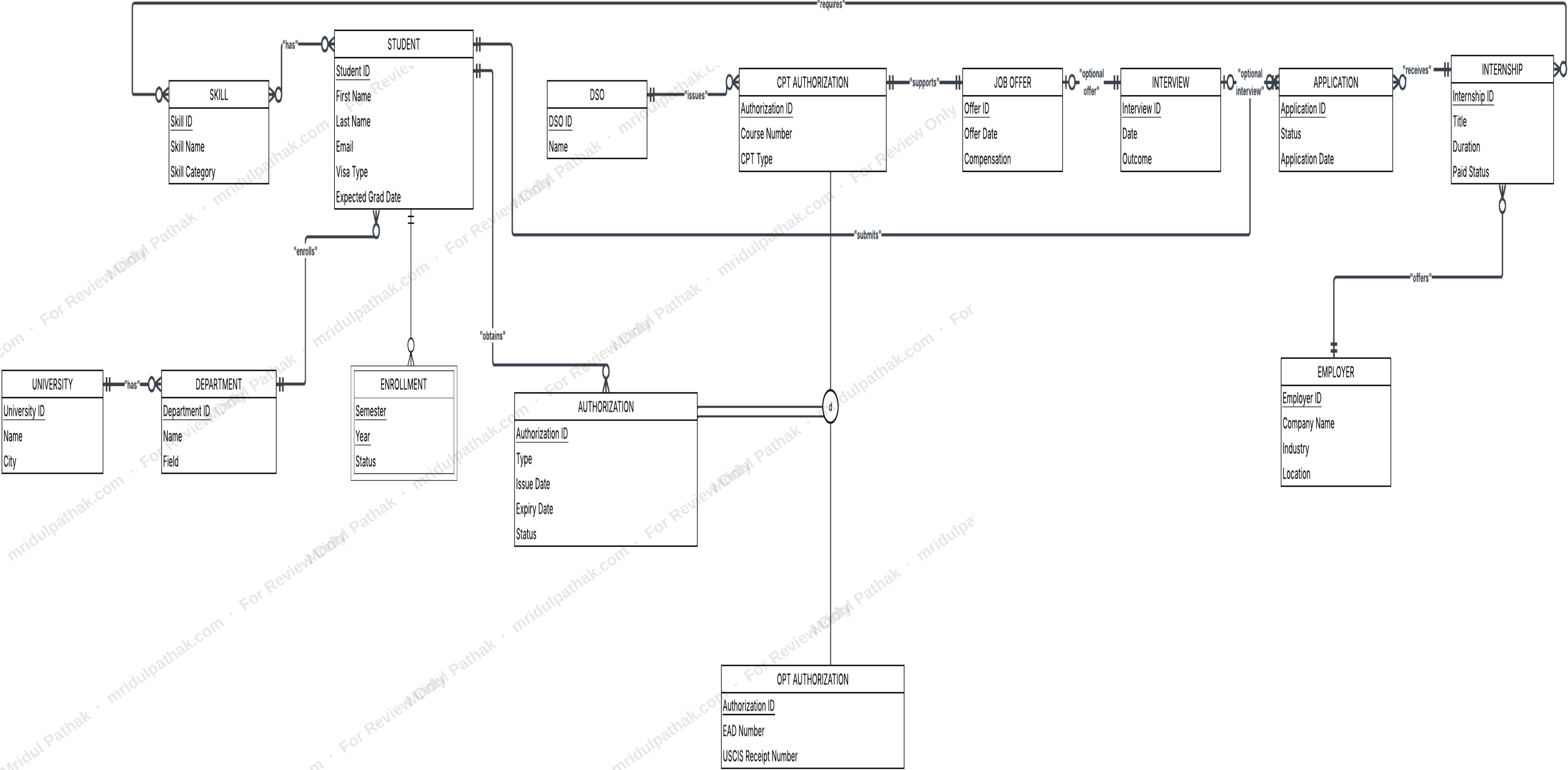
STUDENT_SKILL

UNIVERSITY

The Relational Engine.

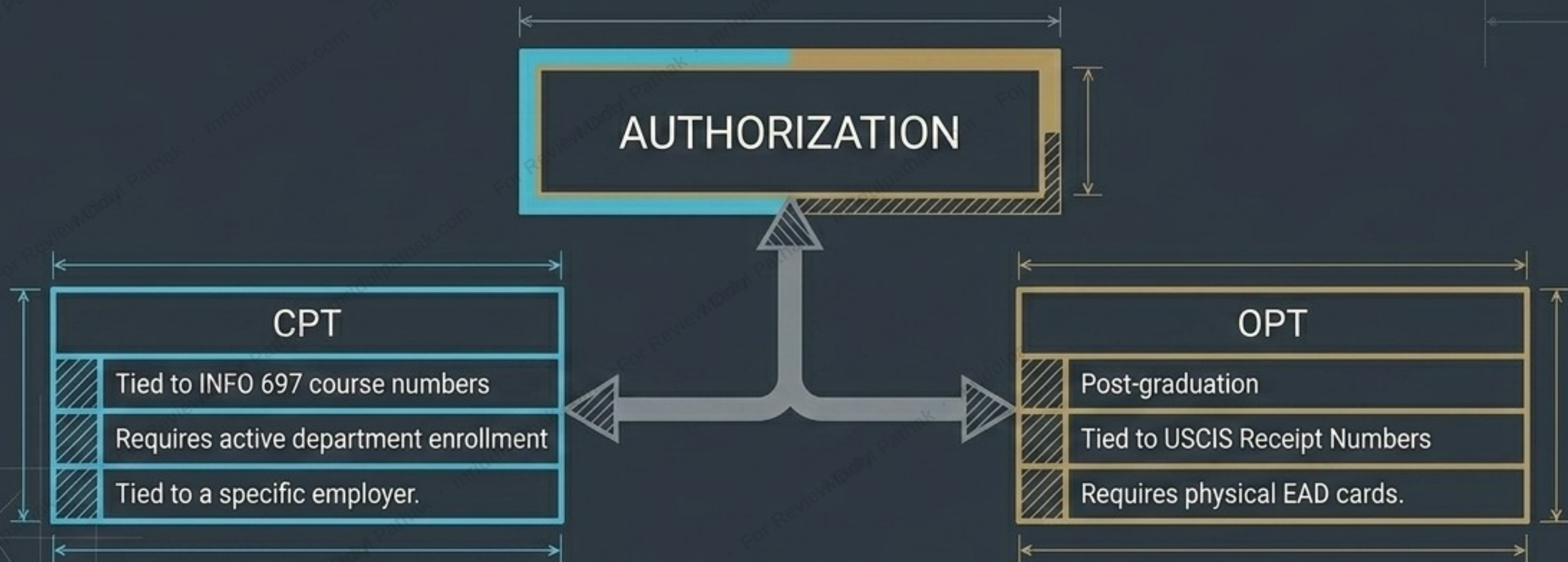
A robust, normalized schema connecting 18+ entities. By separating Skills, Job Offers, and Enrollments into distinct tables, the architecture guarantees data integrity and completely eliminates the redundancies found in traditional spreadsheets.



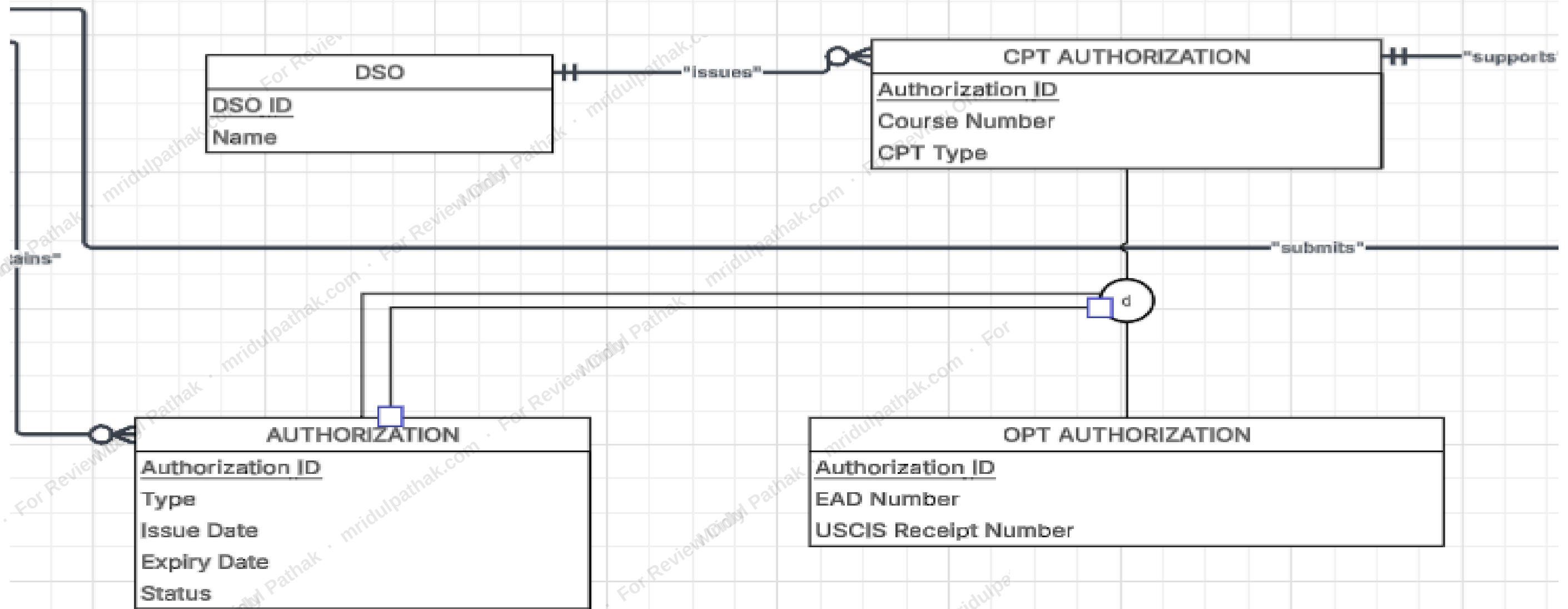


The Disjoint Design: CPT vs. OPT.

Federal law dictates a student cannot hold a single authorization that is simultaneously both CPT and OPT. The database enforces this reality via a Disjoint Subtype Constraint, ensuring zero data anomalies.



⊗ Disjoint Subtype Constraint: {CPT, OPT} are mutually exclusive subtypes of AUTHORIZATION. No overlap permitted.



Instant Retrieval Over Endless Emails.

Query 1 instantaneously returns all active CPT Authorizations. What previously took a student a week of email follow-ups with a DSO is now accessible in 0.02 seconds.

Student ID: 1
Pathak, Mridul
Computer Science
Issue: 03/01/2026
Exp: 08/31/2026

APPROVED

Student ID	Student Name	Department	Auth ID	Issue Date	Expiry Date	Issuing DSO
5	Aisha Muhammed	Information Systems	5	03/20/2026	09/20/2026	Lisa Martinez
3	Priys Patel	Data Analytics	2	03/10/2026	09/10/2026	Emily Davis
1	Mridul Pathak	Computer Science	1	03/01/2026	08/31/2026	Sarah Johnson

3 rows returned in 0.02 seconds

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SQL Commands

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LanguageSQL

Rows10

Clear Command

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Run

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15

16

Query 1: Active CPT Authorizations by Student

SELECT

STUDENT.StudentID AS "Student ID",

STUDENT.FirstName || ' ' || STUDENT.LastName AS "Student Name",

DEPARTMENT.DepartmentName AS "Department",

AUTHORIZATION.AuthorizationID AS "Auth ID",

AUTHORIZATION.IssueDate AS "Issue Date",

AUTHORIZATION.ExpiryDate AS "Expiry Date",

DSO.DSOName AS "Issuing DSO"

FROM STUDENT

JOIN AUTHORIZATION ON STUDENT.StudentID = AUTHORIZATION.StudentID

JOIN CPT_AUTHORIZATION ON AUTHORIZATION.AuthorizationID = CPT_AUTHORIZATION.AuthorizationID

JOIN DSO ON CPT_AUTHORIZATION.DSOID = DSO.DSOID

JOIN DEPARTMENT ON STUDENT.DepartmentID = DEPARTMENT.DepartmentID

WHERE AUTHORIZATION.Status = 'Approved'

ORDER BY AUTHORIZATION.ExpiryDate DESC;

Results

Explain

Describe

Saved SQL

History

3 rows returned in 0.02 secondsDownload

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1

Query 2: Internship Popularity by Application Count

2

SELECT

3

INTERNSHIP.InternshipID AS "Internship ID",

4

INTERNSHIP.Title AS "Internship Title",

5

EMPLOYER.CompanyName AS "Employer",

6

EMPLOYER.Industry AS "Industry",

7

COUNT(APPLICATION.ApplicationID) AS "Total Applications",

8

SUM(CASE WHEN APPLICATION.Status = 'Accepted' THEN 1 ELSE 0 END) AS "Accepted",

9

SUM(CASE WHEN APPLICATION.Status = 'Shortlisted' THEN 1 ELSE 0 END) AS "Shortlisted"

10

FROM INTERNSHIP

11

JOIN EMPLOYER ON INTERNSHIP.EmployerID = EMPLOYER.EmployerID

12

LEFT OUTER JOIN APPLICATION ON INTERNSHIP.InternshipID = APPLICATION.InternshipID

13

GROUP BY INTERNSHIP.InternshipID, INTERNSHIP.Title, EMPLOYER.CompanyName, EMPLOYER.Industry

14

ORDER BY COUNT(APPLICATION.ApplicationID) DESC;

Results

Explain

Describe

Saved SQL

History

Internship ID	Internship Title	Employer	Industry	Total Applications	Accepted	Shortlisted
1	Software Engineering Intern	Google LLC	Technology	2	2	0
2	Data Analytics Intern	Deloitte	Consulting	2	0	2
5	Business Strategy Intern	Deloitte Consulting	Consulting	1	0	0

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Ecosystem Intelligence.

Because the data is structured relationally, administrators gain real-time visibility. Query 4 instantly aggregates total authorizations, pending approvals, and CPT/OPT splits by University—turning administrative bottlenecks into strategic dashboards.

The screenshot displays the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' panel on the left shows a query titled 'Query 4: Authorization Count and Approval Rate by University'. The query is as follows:

```
1 --- Query 4: Authorization Count and Approval Rate by University
2 SELECT
3     UNIVERSITY.UniversityName AS 'University',
4     COUNT(SCHEMARELATED.AuthorizationID) AS 'Total Authorizations',
5     SUM(CASE WHEN AUTHORIZATION.STATUS = 'OPT' THEN 1 ELSE 0 END) AS 'OPT Count',
6     SUM(CASE WHEN AUTHORIZATION.STATUS = 'CPT' THEN 1 ELSE 0 END) AS 'CPT Count',
7     SUM(CASE WHEN AUTHORIZATION.STATUS = 'Approved' THEN 1 ELSE 0 END) AS 'Approved',
8     SUM(CASE WHEN AUTHORIZATION.STATUS = 'Pending' THEN 1 ELSE 0 END) AS 'Pending'
9 FROM UNIVERSITY
10 JOIN SCHEMARELATED ON UNIVERSITY.UniversityID = SCHEMARELATED.UniversityID
11 JOIN STUDENT ON SCHEMARELATED.StudentID = STUDENT.StudentID
12 JOIN AUTHORIZATION ON SCHEMARELATED.AuthorizationID = AUTHORIZATION.AuthorizationID
13 GROUP BY UNIVERSITY.UniversityName
14 ORDER BY COUNT(SCHEMARELATED.AuthorizationID) DESC;
```

The 'Results' panel at the bottom shows the output of the query. The table has the following structure:

University	Total Authorizations	CPT Count	OPT Count	Approved	Pending
Virginia Commonwealth University	2	1	1	2	0
University of Virginia	1	0	1	0	1
Virginia Tech	1	1	0	1	0

The interface also includes a 'Run' button and a 'Save' button. The bottom status bar shows 'Copyright © 1999-2021, Oracle and/or its affiliates. Application Express 21.10'.


```
1  --- Query 4: Authorization Count and Approval Rate by University
2  SELECT
3      UNIVERSITY.UniversityName AS "University",
4      COUNT(AUTHORIZATION.AuthorizationID) AS "Total Authorizations",
5      SUM(CASE WHEN AUTHORIZATION.AuthType = 'CPT' THEN 1 ELSE 0 END) AS "CPT Count",
6      SUM(CASE WHEN AUTHORIZATION.AuthType = 'OPT' THEN 1 ELSE 0 END) AS "OPT Count",
7      SUM(CASE WHEN AUTHORIZATION.Status = 'Approved' THEN 1 ELSE 0 END) AS "Approved",
8      SUM(CASE WHEN AUTHORIZATION.Status = 'Pending' THEN 1 ELSE 0 END) AS "Pending"
9  FROM UNIVERSITY
10 JOIN DEPARTMENT ON UNIVERSITY.UniversityID = DEPARTMENT.UniversityID
11 JOIN STUDENT ON DEPARTMENT.DepartmentID = STUDENT.DepartmentID
12 JOIN AUTHORIZATION ON STUDENT.StudentID = AUTHORIZATION.StudentID
13 GROUP BY UNIVERSITY.UniversityName
14 ORDER BY COUNT(AUTHORIZATION.AuthorizationID) DESC;
```

University	Total Authorizations	CPT Count	OPT Count	Approved	Pending
Virginia Commonwealth University	2	1	1	2	0
University of Virginia	1	0	1	0	1
Virginia Tech	1	1	0	1	0

```
1  --- Query 5: Employer Subsidiary Hierarchy using Unary Self-Join
2  SELECT
3      PARENT.CompanyName AS "Parent Company",
4      PARENT.Industry AS "Industry",
5      SUBSIDIARY.CompanyName AS "Subsidiary",
6      SUBSIDIARY.City || ', ' || SUBSIDIARY.State AS "Subsidiary Location",
7      COUNT(INTERNSHIP.InternshipID) AS "Internships Posted"
8  FROM EMPLOYER PARENT
9  JOIN EMPLOYER SUBSIDIARY ON SUBSIDIARY.ParentEmployerID = PARENT.EmployerID
10 LEFT OUTER JOIN INTERNSHIP ON SUBSIDIARY.EmployerID = INTERNSHIP.EmployerID
11 GROUP BY PARENT.CompanyName, PARENT.Industry, SUBSIDIARY.CompanyName, SUBSIDIARY.City, SUBSIDIARY.State
12 ORDER BY PARENT.CompanyName;
```

Parent Company	Industry	Subsidiary	Subsidiary Location	Internships Posted
Deloitte	Consulting	Deloitte Consulting	Arlington, VA	1
Google LLC	Technology	Google Cloud	Reston, VA	1

2 rows returned in 0.04 seconds Download

↑ SQL Commands

Schema PATHAKM3

Language SQL Rows 10 Clear Command Find Tables Save Run

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1 --- Query 3: Skill Demand vs Student Supply using WITH clause

2 WITH InternshipDemand AS (

3 SELECT

4 SKILL.SkillID,

5 SKILL.SkillName,

6 SKILL.SkillCategory,

7 COUNT(INTERSHIP_SKILL.InternshipID) AS Demand

8 FROM SKILL

9 LEFT OUTER JOIN INTERSHIP_SKILL ON SKILL.SkillID = INTERSHIP_SKILL.SkillID

10 GROUP BY SKILL.SkillID, SKILL.SkillName, SKILL.SkillCategory),

11 StudentSupply AS (

12 SELECT

13 SkillID,

14 COUNT(StudentID) AS Supply

15 FROM STUDENT_SKILL

16 GROUP BY SkillID)

17 SELECT

18 InternshipDemand.SkillName AS "Skill",

19 InternshipDemand.SkillCategory AS "Category",

20 InternshipDemand.Demand AS "Internships Requiring",

21 NVL(StudentSupply.Supply, 0) AS "Students With Skill",

Results Explain Describe Saved SQL History

SQL	Technical	2	1	High Demand
Financial Modeling	Analytical	1	2	Sufficient
Project Management	Management	1	1	Sufficient
Data Analysis	Analytical	1	2	Sufficient

7 rows returned in 0.00 seconds Download

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